then a smooth switch to ⅓ at a density of about 10^7 grams per cubic centimeter when the erratic speeds of the electrons near the speed of light.

The transition from white-dwarf matter to neutron-star matter begins at a density of 4×10^{11} grams per cubic centimeter, according to the Harrison–Wheeler calculations. The calculations show several phases to the transition: In the first phase, the electrons begin to be squeezed into the atomic nuclei, and the nuclei's protons swallow them to form neutrons. The matter, having thereby lost some of its pressure-sustaining electrons, suddenly becomes much less resistant to compression; this causes the sharp cliff in the equation of state (see diagram above). As this first phase proceeds and the resistance plunges, the atomic nuclei become more and more bloated with neutrons, thereby triggering the second phase: Neutrons begin to drip out of (get squeezed out of) the nuclei and into the space between them, alongside the few remaining electrons. These dripped-out neutrons, like the electrons, protest the continuing squeeze with a degeneracy pressure of their own. This neutron degeneracy pressure terminates the over-the-cliff plunge in the equation of state; the resistance to compression recovers and starts rising. In the third phase, at densities between about 10^{12} and 4×10^{12} grams per cubic centimeter, each neutron-bloated nucleus completely disintegrates, that is, breaks up into individual neutrons, forming the neutron gas studied by Oppenheimer and Volkoff, plus a tiny smattering of electrons and protons. From there on upward in density, the equation of state takes on the Oppenheimer–Volkoff neutron-star form (dashed curve in the diagram when nuclear forces are ignored; solid curve using the best 1990s understanding of the influence of nuclear forces).

The results of Wakano's calculations are shown in Figure 5.5. *This figure is the firm and final catalog of cold, dead objects; it answers all the questions we raised, early in this chapter, in our discussion of Figure 5.3.*

In Figure 5.5, the circumference of a star is plotted rightward and its mass upward. Any star with circumference and mass in the white region of the figure has a stronger internal gravity than its pressure, so its gravity makes the star shrink leftward in the diagram. Any star in the shaded region has a stronger pressure than gravity, so its pressure makes the star expand